The Constrained Continuum Emergence Framework: Ontology and Field Equations

Allan Gibson

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Abstract

The Constrained Continuum Emergence Framework (CCEF) describes gravity, gauge interactions, and matter as collective properties of a single continuum order-parameter field defined on a fixed background manifold. The metric, the Einstein–Hilbert action, the graviton, an abelian gauge field, and localized matter all appear as induced responses and topological excitations of this field. The observable universe corresponds to the disordered (symmetric) phase of the order parameter, in which the emergent abelian sector is in its Coulomb regime and the associated gauge boson is massless. This paper states the ontology of the framework and collects its defining equations: the order-parameter action, the induced gravitational sector, the composite graviton, the phase structure of the emergent abelian field, the matter sector, the cosmological equations, and the fixed parameter set.

1 Ontology

CCEF postulates a single fundamental entity: a continuum order-parameter field Φ defined on a fixed, non-dynamical four-manifold with a flat reference structure. Every observed structure is a property of Φ:

1. **Geometry is not fundamental.** The spacetime metric $g_{\mu\nu}$ is the source term conjugate to the field's stress tensor $T_{\mu\nu}$; it enters as a coupling, not as an independent degree of freedom.
2. **Gravity is an induced response.** Fluctuations of Φ generate an Einstein–Hilbert term for $g_{\mu\nu}$; Newton's constant is fixed by the field content and by the scale M at which the field's response becomes non-local.
3. **The graviton is composite.** The spin-two excitation is the transverse–traceless channel of the connected two-point function of $T_{\mu\nu}$. Conservation of $T_{\mu\nu}$ protects it at zero mass.
4. **Gauge fields are collective.** An abelian connection is built from the phase of Φ; its dynamics follow from the four-derivative sector of the order-parameter action.
5. **Matter is topological.** Localized, particle-like states are finite-energy solitons of Φ, classified by the homotopy of its target space.
6. **Phases.** Φ admits a disordered (symmetric) phase and an ordered (broken) phase. The observable universe is the disordered phase. The two phases carry different low-energy content, in particular a massless versus a massive emergent vector boson.

The remainder of the paper makes these statements quantitative.

2 Field content

The order parameter takes values in complex projective space $\mathbb{CP}^{N-1}$. It is represented by an N-component complex field $z = (z_1, \dots, z_N)$ subject to

$$z^\dagger z = 1, \qquad z \sim e^{i\alpha} z, \tag{1}$$

so that the physical target is $\mathbb{CP}^{N-1} = S^{2N-1}/U(1)$. The phase redundancy defines a composite abelian connection

$$a_\mu \;=\; -\,i\, z^\dagger \partial_\mu z, \qquad f_{\mu\nu} \;=\; \partial_\mu a_\nu - \partial_\nu a_\mu, \tag{2}$$

and the gauge-covariant derivative $D_\mu z = (\partial_\mu - i a_\mu) z$.

Gravitational sector. The gravitational sector uses $N = 2$, for which $\mathbb{CP}^1 \cong S^2$. It is equivalently described by a unit three-vector

$$\boldsymbol{n} = z^\dagger \boldsymbol{\sigma}\, z, \qquad \boldsymbol{n} \cdot \boldsymbol{n} = 1, \tag{3}$$

with $\boldsymbol{\sigma}$ the Pauli matrices. In this description the abelian field strength of (2) is

$$f_{\mu\nu} \;=\; \boldsymbol{n} \cdot (\partial_\mu \boldsymbol{n} \times \partial_\nu \boldsymbol{n})\,. \tag{4}$$

Matter sector. The matter sector uses an $SU(2) \cong S^3$-valued field $U(x)$, carrying the topological charges that label localized states (Section 7).

3 Order-parameter action

The action is organized as an expansion in derivatives, with a two-derivative term, a marginal four-derivative sector suppressed by the non-locality scale M, and a potential:

$$S[\boldsymbol{n}] \;=\; \int \mathrm{d}^4 x\, \Big[\mathcal{L}_2 + \mathcal{L}_4 + \mathcal{L}_V \Big]. \tag{5}$$

Two-derivative term.

$$\mathcal{L}_2 \;=\; \frac{1}{2g^2}\, \partial_\mu \boldsymbol{n} \cdot \partial^\mu \boldsymbol{n}. \tag{6}$$

Four-derivative (marginal) sector. Define the independent quartic invariants

$$\mathcal{O}_1 = \big(\partial_\mu \boldsymbol{n} \cdot \partial^\mu \boldsymbol{n}\big)^2, \qquad \mathcal{O}_2 = \big(\partial_\mu \boldsymbol{n} \cdot \partial_\nu \boldsymbol{n}\big)\big(\partial^\mu \boldsymbol{n} \cdot \partial^\nu \boldsymbol{n}\big), \qquad \mathcal{O}_4 = (\Box \boldsymbol{n}) \cdot (\Box \boldsymbol{n}). \tag{7}$$

The algebraic identity $f_{\mu\nu} f^{\mu\nu} = \mathcal{O}_1 - \mathcal{O}_2$ identifies the antisymmetric quartic invariant with the square of the emergent field strength (4). The marginal sector is

$$\mathcal{L}_4 \;=\; \frac{1}{M^2}\Big[A_2\, (\mathcal{O}_1 - \mathcal{O}_2) \;+\; A_3\, \mathcal{O}_4 \Big] \;=\; \frac{A_2}{M^2}\, f_{\mu\nu} f^{\mu\nu} \;+\; \frac{A_3}{M^2}\, (\Box \boldsymbol{n})^2. \tag{8}$$

The coefficient A_2 sets the coupling of the emergent abelian field, $e_B^{-2} = 4A_2$.

Potential. The potential fixes the phase structure and the mass gap:

$$\mathcal{L}_V \;=\; -\,\tfrac{1}{2} A_1 E_0^2\, \big(1 - n_3^2\big) \;-\; \tfrac{1}{4} A_4 E_0^2\, \mathcal{C}(\boldsymbol{n}), \tag{9}$$

with E_0 the intrinsic energy unit of the framework, A_1 the anisotropy coefficient, $\mathcal{C}(\boldsymbol{n})$ the constraint term, and A_4 setting the gap $\Lambda = \sqrt{A_4}\, E_0$.

4 Induced gravitational sector

Integrating over order-parameter fluctuations generates a local effective action for the metric that couples to $T_{\mu\nu}$. Its curvature expansion is

$$\Gamma_{\rm grav}[g] \;=\; \int d^4x\,\sqrt{-g}\,\Big[\,-\,\frac{\Lambda_{\rm eff}}{8\pi G}\;+\;\frac{R}{16\pi G}\;+\;\alpha_2\,R^2\;+\;\alpha_3\,R_{\mu\nu}R^{\mu\nu}\;+\;\cdots\Big]. \tag{10}$$

The coefficient of the Einstein–Hilbert term is fixed by the field content and the non-locality scale M,

$$\frac{1}{16\pi G}\;=\;\frac{N_{\rm eff}}{6}\cdot\frac{M^2}{16\pi^2}, \qquad\Longleftrightarrow\qquad M_{\rm Pl}^2\;=\;\frac{N_{\rm eff}\,M^2}{48\pi^2}, \tag{11}$$

with $N_{\rm eff}$ the effective number of contributing modes. The sign of the leading heat-kernel coefficient,

$$a_1\;=\;\frac{N_{\rm eff}}{6}\,R\;>\;0, \tag{12}$$

fixes the positivity of the graviton kinetic term. Relation (11) places the non-locality scale at $M \simeq 1.8\times10^{19}\,\mathrm{GeV}$ for $N_{\rm eff}\simeq 9$.

5 Composite graviton

The spin-two excitation is the transverse–traceless channel of the connected stress-tensor correlator,

$$h_{\mu\nu}(x)\;\longleftrightarrow\;\big\langle T_{\mu\nu}(x)\,T_{\rho\sigma}(0)\big\rangle^{\rm TT}\;\equiv\;\Pi^{\rm TT}_{\mu\nu,\rho\sigma}(x). \tag{13}$$

Conservation of the stress tensor,

$$\partial^\mu T_{\mu\nu}=0 \qquad\Longrightarrow\qquad p^\mu\,\Pi_{\mu\nu,\rho\sigma}(p)=0, \tag{14}$$

is the Ward identity of the emergent diffeomorphism invariance. It fixes the transverse–traceless self-energy to a massless pole,

$$\Pi^{\rm TT}(p)\;=\;\frac{Z_{\rm TT}}{p^2}+(\text{regular}),\qquad Z_{\rm TT}>0,\qquad m_{\rm graviton}=0. \tag{15}$$

The pole persists in the disordered phase, where the spectrum carries a gap $m_{\rm gap}\sim\Lambda$ and the pole sits below the two-particle continuum at $p^2=4\,m_{\rm gap}^2$.

6 Phase structure and the emergent abelian field

The order parameter admits two phases distinguished by the condensate $\langle|\Phi|\rangle$:

Disordered (symmetric) phase. $\langle|\Phi|\rangle=0$. The emergent abelian field is in the Coulomb regime; the composite connection a_μ of (2) propagates as a massless vector with coupling

$$e_B^2\;=\;\frac{1}{4A_2}\;\approx\;0.765. \tag{16}$$

This is the phase of the observable universe, and accounts for a massless gauge boson.

Ordered (broken) phase. $\langle|\Phi|\rangle=1$. The abelian symmetry is realized in the Higgs regime; the composite vector acquires a mass

$$m_\rho\;=\;\sqrt{\frac{A_1}{2A_2}}\,E_0\;\approx\;152\,\mathrm{MeV}. \tag{17}$$

The transition between the two phases is second order in the effective potential, with the condensate serving as the order parameter.

7 Matter sector

Localized states are finite-energy topological solitons of the order-parameter field. The relevant homotopy groups classify the charges:

$$\pi_3(S^2) = \mathbb{Z} \qquad \text{(Hopf charge, } \boldsymbol{n} \text{ sector)}, \tag{18}$$

$$\pi_3(S^3) = \mathbb{Z} \qquad \text{(winding number, } SU(2) \text{ sector)}. \tag{19}$$

The energy functional of the $\boldsymbol{n}$ sector supporting these states combines the two- and four-derivative terms of Section 3,

$$E[\boldsymbol{n}] \;=\; \int \mathrm{d}^3x \,\Big[\, \frac{1}{2g^2}\, \partial_i \boldsymbol{n} \cdot \partial_i \boldsymbol{n} \;+\; \frac{A_2}{M^2}\, f_{ij} f_{ij} \,\Big], \tag{20}$$

whose stationary configurations at fixed Hopf charge define the soliton spectrum. The soliton mass scale is set by E_0.

8 Cosmological equations

The induced gravitational dynamics take the Einstein form,

$$G_{\mu\nu} \;+\; \Lambda_{\mathrm{eff}}\, g_{\mu\nu} \;=\; 8\pi G\, T_{\mu\nu}, \tag{21}$$

with $T_{\mu\nu}$ the stress tensor of the disordered-phase order parameter. For a homogeneous, isotropic background with metric $\mathrm{d}s^2 = -\mathrm{d}t^2 + a(t)^2\, \mathrm{d}\boldsymbol{x}^2$, the field equations reduce to

$$\left(\frac{\dot a}{a}\right)^2 = \frac{8\pi G}{3}\,\rho \;+\; \frac{\Lambda_{\mathrm{eff}}}{3}, \qquad \frac{\ddot a}{a} = -\frac{4\pi G}{3}\,(\rho + 3p) \;+\; \frac{\Lambda_{\mathrm{eff}}}{3}, \tag{22}$$

with ρ and p the energy density and pressure of the order-parameter field. The effective cosmological constant takes the value $\Lambda_{\mathrm{eff}} \simeq 2.5 \times 10^{-47}\,\mathrm{GeV}^4$.

9 Scalar spectrum

The primordial scalar spectrum is set by the two-point function of the order-parameter stress tensor evaluated at the critical point of the disordered phase. Stress-tensor conservation (14) fixes $T_{\mu\nu}$ at scaling dimension $d = 3$, so the leading spectrum is scale-invariant,

$$n_s\big|_{\text{fixed point}} = 1. \tag{23}$$

The departure from scale invariance is controlled by the anomalous dimension η of the critical order parameter (three-dimensional $O(3)$ universality class),

$$n_s \;=\; 1 - \eta, \qquad \eta \simeq 0.0375 \quad\Longrightarrow\quad n_s \simeq 0.963. \tag{24}$$

The tilt is red, with the same conservation law (14) controlling both the flatness of the spectrum and the masslessness of the graviton.

10 Parameters

Table 1 lists the fixed parameters of the framework.

Symbol	Value	Role
E_0	$122.5\,\mathrm{MeV}$	intrinsic energy unit
$\Lambda = \sqrt{A_4}\,E_0$	$231\,\mathrm{MeV}$	mass gap
A_1	1	anisotropy coefficient
A_2	0.3268	marginal (abelian) coefficient
A_4	3.555	gap coefficient
$e_B^2 = 1/4A_2$	0.765	emergent abelian coupling
m_ρ	$152\,\mathrm{MeV}$	vector mass (ordered phase)
M	$1.77 \times 10^{19}\,\mathrm{GeV}$	non-locality scale
M_{Pl}	$2.435 \times 10^{18}\,\mathrm{GeV}$	reduced Planck mass
N_{eff}	$\simeq 9$	effective mode number
Λ_{eff}	$2.5 \times 10^{-47}\,\mathrm{GeV}^4$	effective cosmological constant
$\ln(M/\Lambda)$	45.8	scale hierarchy (logarithmic)
$(\Lambda/M)^2$	1.7×10^{-40}	scale hierarchy (ratio)
η	0.0375	$O(3)$ anomalous dimension

Table 1: Fixed parameters of the framework.

Summary

The framework specifies a single continuum order-parameter field valued in $\mathbb{CP}^{N-1}$, an action with a two-derivative term, a marginal four-derivative sector at the scale M, and a potential. From this content the metric acquires an Einstein–Hilbert action with Newton constant fixed by (11); the spin-two excitation is the transverse–traceless stress-tensor channel, massless by (14); an emergent abelian field is massless in the disordered phase and massive in the ordered phase; matter states are topological solitons; and the cosmological dynamics take the Einstein form with an effective cosmological constant. The scalar spectrum is scale-invariant at leading order with a red tilt $n_s = 1 - \eta$.